Evaluation of diagnostic yield, laboratory data and clinical correlates in an established HFE genetic testing service

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INTRODUCTION

Advances in laboratory information systems now enable integration of genetic, biochemical and clinical data at unprecedented scale, creating opportunities to evaluate diagnostic practices in real-world, locally served populations. Using 25 years of HFE testing at St James's Hospital, we examined how testing patterns, biochemical profiles and reported clinical features relate to diagnostic yield and HFE genotype.

AIM | Evaluate long-term testing performance and identify biochemical and clinical features associated with genotype positivity, particularly C282Y homozygosity.

METHODS | Secure integration of genetics, biochemistry and referral data

SHIRE GENETIC DATABASE

2004-2024
27,567 requests
24,866 patients

Genotype, demographics, referral location and documented clinical features.

CENTRALISED LIMS | BIOCHEMISTRY + HAEMATOLOGY [HAEMATINICS]

1 Oct 2020–30 Sep 2024
361,082 unique complete-identity patients | 12,737,369 consolidated, non-duplicate analyte results.

TSAT and ferritin
+ Albumin, total protein, bilirubin, ALP, GGT, AST, ALT

SECURE LINKAGE

SHIRE ↔ LIMS
(Name | DOB | sex | +/- MRN)

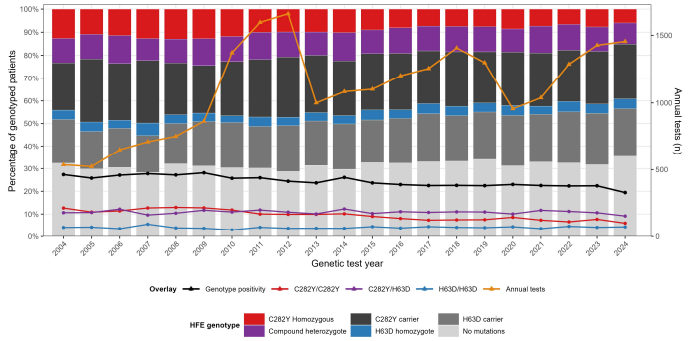
- Pseudonymisation
- 24,260 genotyped patients.
- TSAT+ferritin data (n= 7,030)
- LIMS-enrichment (n=10,346) → +47.2%
- Extended LIMS data on 2,453 genotyped patients (complete)

ANALYSIS

- Diagnostic yield trends
- Ferritin–TSAT vs genotype
- EASL biochemical threshold stratification
- ROC/AUC analysis
- Exploratory modelling: elastic net, biomarker feature importance
- Clinical correlates from documented referral information

RESULTS

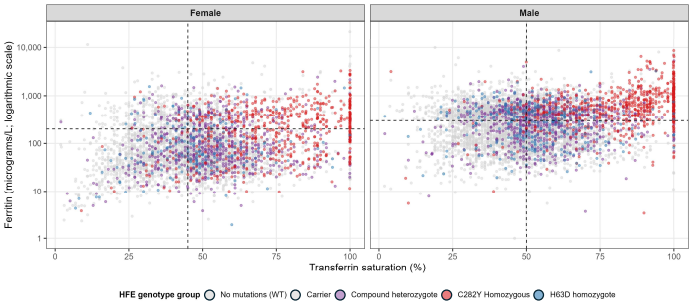
A | Annual testing and diagnostic yield 2004-2024



5,792/24,260 HFE positive (23.9%); diagnostic yield declined (OR/year 0.981, 95% CI 0.976-0.986)
2,215/24,260 C282Y/C282Y + (9.1%); diagnostic yield declined (OR/year 0.962, 95% CI 0.955–0.969)

B | Ferritin and TSAT by genotype

Scatterplot of ferritin (x-axis) versus TSAT (y-axis) in 10,347 unique patients with linked genetic (SHIRE) and biochemical (LIMS) records.

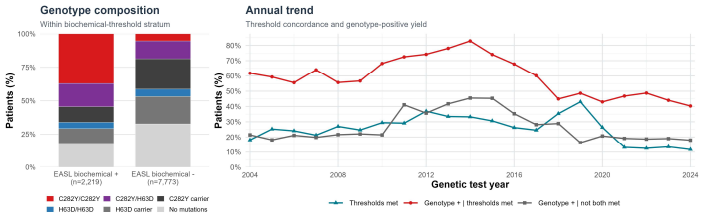


HFE GENOTYPE POSITIVITY BY EASL BIOCHEMICAL THRESHOLD CATEGORY (TSAT, ferritin)

58.6%	33.2%	18.0%	16.2%
both | 1,339 / 2,286	TSAT only | 1,125 / 3,388	ferritin only | 334 / 1,854	neither | 457 / 2,818

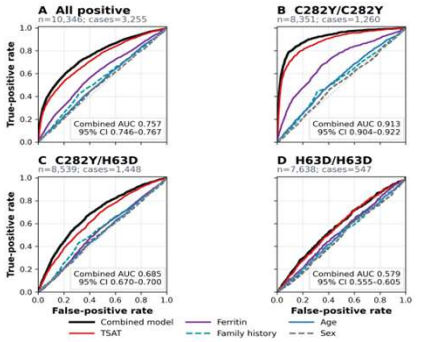
EASL (2022) thresholds: Women, TSAT >45% and ferritin >200 µg/L; men, TSAT >50% and ferritin >300 µg/L

C | Genotype composition by biochemical threshold status



2004-2024 subset: genotype-positive in 59.3% meeting both thresholds vs 24.1% not meeting both.

D | Ferritin, TSAT, age, sex and family history

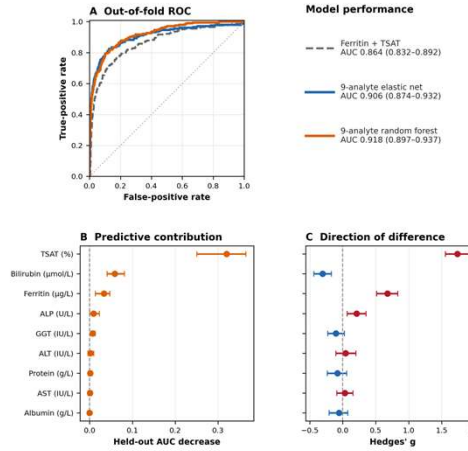


Combined features discriminated C282Y/C282Y particularly well (AUC 0.913), with TSAT the strongest individual predictor.

Performance was lower for C282Y/H63D and approached chance for H63D/H63D.

Combined AUC: all positive 0.757 | C282Y/C282Y 0.913 | C282Y/H63D 0.685 | H63D/H63D 0.579

E | Complete biochemical profiles



2,453 patients | 205 C282Y/C282Y | 12-analyte complete profile; 9 modelled | ≤365 days; 30-day span

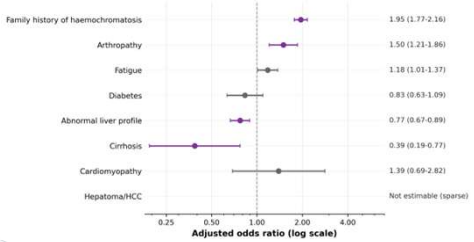
AUC 0.906
9-analyte elastic net vs ferritin + TSAT 0.864

Delta AUC 0.042 (95% CI 0.027–0.057). Wider biochemistry added modest discrimination beyond ferritin and TSAT; TSAT remained dominant.

Outcome: C282Y/C282Y. Comparator: no mutations plus C282Y or H63D carrier states; compound heterozygotes and H63D/H63D excluded.

F | Referral features and C282Y homozygosity

Patients (n = 9,926). Documented referral features: family history (42.9%), abnormal liver profile (11.9%), fatigue (9.9%), arthropathy (3.9%), diabetes (3.4%), cirrhosis (0.7%), cardiomyopathy (0.4%) and hepatoma/HCC (0.1%).



aOR 1.95
Family history | 1.77–2.16

aOR 1.50
Arthropathy | 1.21–1.86

aOR 0.39 (?)
Cirrhosis | 0.19–0.77

CONCLUSIONS

Integration of 25 years of genetic, biochemical and referral data from the HFE testing service at St James's Hospital (SJH) enabled evaluation of 24,260 patients, the majority referred from general practice (>75%). Data linkage increased availability of paired ferritin–TSAT results by 47%. Diagnostic yield declined over time (26.3% to 20.9%), while meeting both EASL biochemical thresholds markedly enriched genotype positivity (58.6%), particularly C282Y homozygosity. TSAT was the strongest biochemical discriminator, with broader biochemical profiles providing modest additional predictive information. Family history and arthropathy were independently associated with C282Y homozygosity.

Interpretation is limited by retrospective, referral-selected data. Future work should examine differences between GP and hospital referrals, evaluate structured requesting, and integrate clinical outcomes and disease characteristics to better assess predictors of genotype positivity and clinical significance in the local population.

ACKNOWLEDGEMENTS / REFERENCES

We thank colleagues in SJH Biochemistry, Haematology, Laboratory IT and data management.

Code: Analyses and figures used Python and R, with PowerShell/PowerPoint for poster preparation. https://github.com/mmacaogain/HFE-service-evaluation
OpenAI Codex assisted with code development, review and figure preparation. AI inputs were restricted to dataset structures, code and non-identifying aggregate results; patient-level data were not shared with AI services

Governance: SJH QSID 103; JREC 5281 (service evaluation)

Reference: EASL Clinical Practice Guidelines. *J Hepatol.* 2022;77:479–502.